PCBC H Parametrics Practice Test C Name _____ _____/12

Non-Calculator | 30 minutes total for all four questions

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

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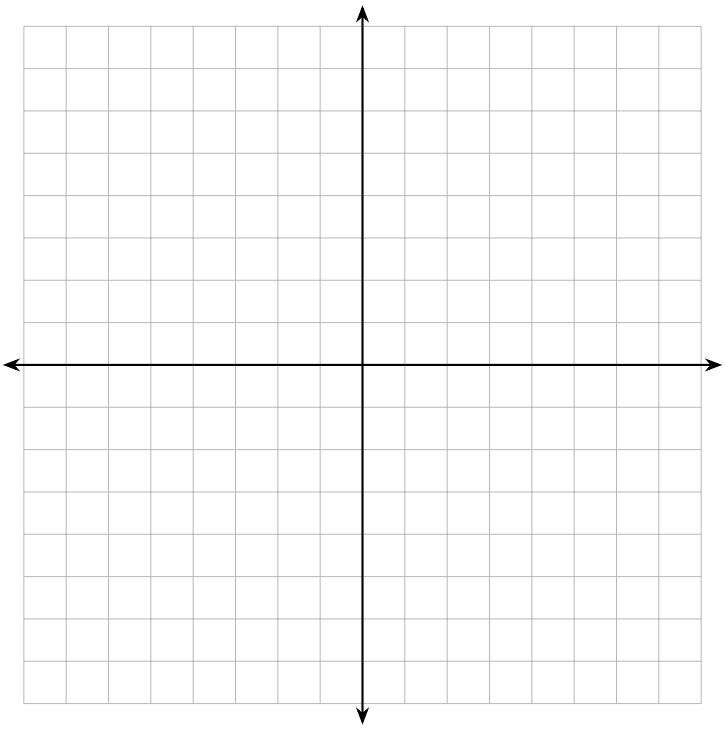
Signature _____

SHOW ALL WORK FOR FULL CREDIT!

Practice set: 3 points per question is a practice split of the posted 12 points.

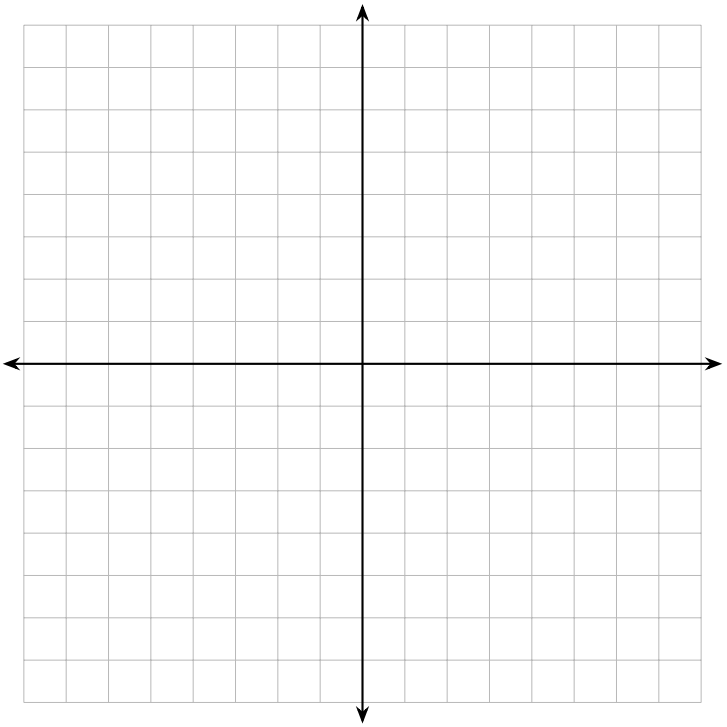
Graph the plane curve defined by the given parametric equations. Then find a rectangular equation for the curve. State any restrictions on the domain and show the direction of the curve. (3 points)

1. $x = 3 - t, \quad y = t^2 - 4t, \quad$ for t in $[0,4]$ **3 pts**



Find a rectangular equation for the given plane curve, and graph the curve. Show the direction of the curve. (3 points)

2. $x = 1 + 2\sin t, \quad y = -2 + 3\cos t, \quad \text{for } t \text{ in } [0, 2\pi]$ **3 pts**



Find two different parametric representations for the equation of the parabola. (3 points)

3. $y = x^2 - 6x + 7$ **3 pts**

Representation 1 *Representation 2*

Calculator | 30 minutes total for all four questions

SHOW ALL WORK FOR FULL CREDIT!

Show your set up (what you are putting into the calculator) and round your answers to the nearest thousandth.
(practice instructions)

4. The path of a projectile follows the parametric equations

$$x = v_0(\cos\theta)t \qquad \text{and} \qquad y = -16t^2 + v_0(\sin\theta)t + h_0$$

where θ is the angle of inclination of the projectile at the launch, v_0 is the initial velocity of the projectile in feet per second, and h_0 is the initial height of the projectile in feet. *(Formulas printed here as a practice choice.)*

A ball is thrown from a height of 5.5 ft at an angle of inclination of 42° with a velocity of 64 ft/sec. Write the parametric equations that model its path. For how many seconds is the ball in flight? What horizontal distance does it travel? **3 pts**

PCBC H Parametrics Practice Test C: Answer Key

12 points

Every value was checked in Python (sympy and 30-digit arithmetic). Work is on the left, the reason for each line is on the right.

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1. $x = 3 - t,\ y = t^2 - 4t,$ for t in $[0, 4]$ *Non-Calculator*

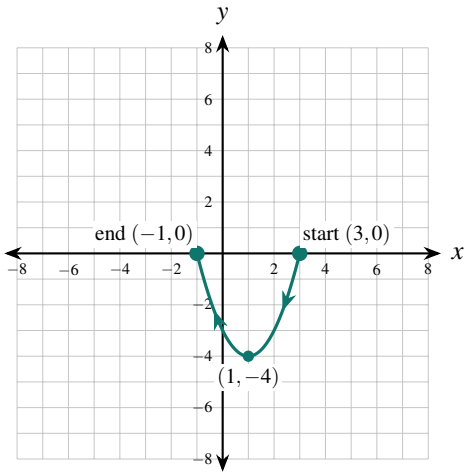
$\boxed{y = x^2 - 2x - 3, \quad -1 \le x \le 3}$

Graph: parabola piece from $(3, 0)$ down to the vertex $(1, -4)$, then up to $(-1, 0)$, traced right to left.

Work	Why
$t = 3 - x$	x gives t in one step.
$y = (3 - x)^2 - 4(3 - x)$	Substitute into y.
$y = 9 - 6x + x^2 - 12 + 4x = x^2 - 2x - 3$	Expand and combine.
$t = 0 \Rightarrow (3, 0);\ t = 4 \Rightarrow (-1, 0)$	Endpoints.
$-1 \le x \le 3$	$x = 3 - t$ only decreases, so the t endpoints give the x endpoints.
$t = 2 \Rightarrow (1, -4)$	y turns at $t = 2$: lowest point, so $-4 \le y \le 0$.
arrows right to left	x gets smaller as t increases.

t	(x, y)
0	$(3, 0)$
1	$(2, -3)$
2	$(1, -4)$
3	$(0, -3)$
4	$(-1, 0)$

Starts on the right. Arrows left to right here would be the most common mistake.



Practice scoring (3 pts): 1 for the equation, 1 for $-1 \le x \le 3$, 1 for the graph with end dots and right-to-left arrows.

Trap: drawing the arrows left to right because the graph "reads" that way. $x = 3 - t$ decreases, so the curve moves left.

2. $x = 1 + 2\sin t,\ y = -2 + 3\cos t,$ for t in $[0, 2\pi]$ *Non-Calculator*

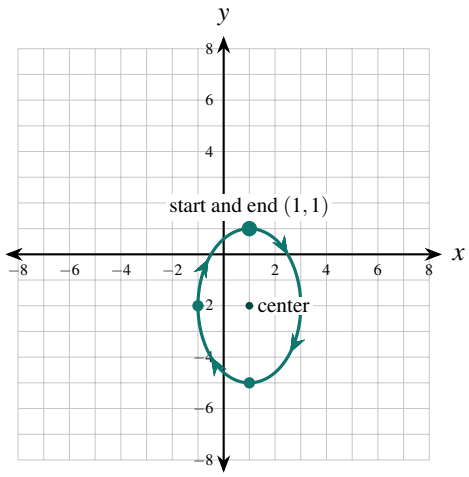
$$\frac{(x-1)^2}{4} + \frac{(y+2)^2}{9} = 1$$

Whole ellipse, center $(1, -2)$, traced once clockwise, starting and ending at $(1, 1)$.

Work	Why
$\sin t = \dfrac{x-1}{2}, \quad \cos t = \dfrac{y+2}{3}$	Isolate each trig function.
$\sin^2 t + \cos^2 t = 1$	Pythagorean identity.
$\left(\dfrac{x-1}{2}\right)^2 + \left(\dfrac{y+2}{3}\right)^2 = 1$	Substitute.
$\dfrac{(x-1)^2}{4} + \dfrac{(y+2)^2}{9} = 1$	Ellipse, center $(1, -2)$, 2 left and right, 3 up and down.
$t = 0 : (1, 1);\ t = \frac{\pi}{2} : (3, -2);\ t = \pi : (1, -5)$	Top, then right, then bottom: clockwise.
$t = \frac{3\pi}{2} : (-1, -2);\ t = 2\pi : (1, 1)$	Back to the start: one full trip.

t	(x, y)
0	$(1, 1)$
$\frac{\pi}{2}$	$(3, -2)$
π	$(1, -5)$
$\frac{3\pi}{2}$	$(-1, -2)$
2π	$(1, 1)$

$t = 0$ and $t = 2\pi$ land on the same point, so there is one start/end point, not two separate end dots.



Practice scoring (3 pts): 1 for the equation, 1 for an ellipse with center $(1, -2)$ drawn to the right size, 1 for clockwise arrows starting at $(1, 1)$.

Trap: assuming counterclockwise. With $\sin t$ in x and $\cos t$ in y (both coefficients positive), the start is the top and the motion is clockwise.

3. $y = x^2 - 6x + 7$ *Non-Calculator*

$\boxed{x = t,\ y = t^2 - 6t + 7,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t + 3,\ y = t^2 - 2,\ \text{for } t \text{ in } (-\infty, \infty)}$

Work	**Why**
Representation 1: $x = t$	Let x be the parameter.
$y = t^2 - 6t + 7,\ \ t$ in $(-\infty, \infty)$	Every real x is allowed, so every real t.
Representation 2: $y = (x^2 - 6x + 9) - 9 + 7$	Complete the square: half of -6 is -3, $(-3)^2 = 9$.
$y = (x - 3)^2 - 2$	Vertex form, vertex $(3, -2)$.
$t = x - 3 \ \Rightarrow\ x = t + 3$	Let t be the inside of the square.
$y = t^2 - 2,\ \ t$ in $(-\infty, \infty)$	Substitute.
Check: $(t+3)^2 - 6(t+3) + 7 = t^2 + 6t + 9 - 6t - 18 + 7 = t^2 - 2$	Put $x = t + 3$ back into the original.

Practice scoring (3 pts): 1 for each correct pair of equations, 1 for the t interval on both. Any other correct pair also counts.

Trap: $t = x - 3$ means $x = t + 3$. Write "for t in $(-\infty, \infty)$" with both.

4. Ball launched from $h_0 = 5.5$ ft at $\theta = 42^\circ$ with $v_0 = 64$ ft/sec. *Calculator*

$\boxed{x = (64\cos 42^\circ)t, \quad y = -16t^2 + (64\sin 42^\circ)t + 5.5}$

In flight $\boxed{\approx 2.799 \text{ sec}}$, travels $\boxed{\approx 133.139 \text{ ft}}$ horizontally.

Work	**Why**
$x = (64\cos 42^\circ)t \approx 47.561t$	Horizontal: $v_0 \cos\theta$ times t.
$y = -16t^2 + (64\sin 42^\circ)t + 5.5 \approx -16t^2 + 42.824t + 5.5$	Vertical: gravity, $v_0 \sin\theta$ times t, plus the starting height.
$0 = -16t^2 + (64\sin 42^\circ)t + 5.5$	It lands when $y = 0$.
$t = \dfrac{-64\sin 42^\circ \pm \sqrt{(64\sin 42^\circ)^2 - 4(-16)(5.5)}}{2(-16)}$	Quadratic formula with $a = -16$, $b = 64\sin 42^\circ$, $c = 5.5$.
$t = \dfrac{-42.824 \pm \sqrt{2185.926}}{-32}$	$b^2 \approx 1833.926$ and $-4ac = +352$.
$t \approx 2.799$ or $t \approx -0.123$	Reject the negative time.
$x = (64\cos 42^\circ)(2.79932\ldots)$	Put the unrounded t into x (keep it stored).
$x \approx 133.139$ ft	Round only at the end.

Calculator setup (TI-Nspire), and a graph check

- Angle mode: **Degree**, or type the $^\circ$ symbol after the angle. In radian mode every number is wrong.
- Graph check: Graphs page, Menu, Graph Entry/Edit, Parametric. $x1(t) = 64\cos(42^\circ)t$, $y1(t) = -16t^2 + 64\sin(42^\circ)t + 5.5$, $0 \le t \le 3$. Window x: 0 to 150, y: -5 to 40. Trace near $t = 2.799$: y crosses 0 with $x \approx 133.1$.
- Not asked: the top of the path is at $t = \frac{64\sin 42^\circ}{32} \approx 1.338$ sec, height ≈ 34.155 ft.

Practice scoring (3 pts): 1 for both parametric equations, 1 for the flight time, 1 for the horizontal distance.

Trap: rounding t to 2.80 before finding x gives ≈ 133.172 ft, which is off at the thousandths. Keep the full t.